

Pump-Current Conversions in the `twpa_jax` Maps

Thesis working note

18 July 2026

1 Purpose

The gain-map programs convert an external pump-power coordinate into an on-chip current before solving the nonlinear IPM circuit. Two different conventions have been used historically through the option `-pump-current-jc-scale`. They must not be confused with the ordinary RMS-to-peak or Fourier-coefficient conversions.

2 Power to physical peak current

For an external pump power P_{dBm} , after subtracting the line attenuation A_{dB} , the on-chip power is

$$P_W = 10^{-3} 10^{(P_{\text{dBm}} - A_{\text{dB}})/10}. \quad (1)$$

The map code then computes

$$I_{\text{peak}} = \sqrt{\frac{2P_W}{Z_0}}. \quad (2)$$

Thus the current stored in each map point is already a peak current. In particular, multiplying it by $\sqrt{2}$ again would be incorrect.

3 Fourier and phasor bookkeeping

For a real single-tone current, the conventions are

$$i(t) = I_{\text{peak}} \cos(\omega t + \phi), \quad (3)$$

$$= 2 \operatorname{Re} [I_+ e^{i\omega t}], \quad (4)$$

$$I_+ = \frac{I_{\text{peak}} e^{i\phi}}{2}. \quad (5)$$

Equivalently, with an RMS RF phasor I_{rms} ,

$$i(t) = \sqrt{2} \operatorname{Re} [I_{\text{rms}} e^{i\omega t}], \quad I_{\text{rms}} = \frac{I_{\text{peak}}}{\sqrt{2}}, \quad I_+ = \frac{I_{\text{rms}}}{\sqrt{2}}. \quad (6)$$

The positive-phasor pump basis reconstructs the real waveform as $2 \operatorname{Re} \sum_k X_k e^{ik\omega_p t}$. This factor of two belongs to the reconstruction of a positive-frequency coefficient; it is not an instruction to multiply the already-computed map current by two.

Setting	Source current	Meaning	Use
$s_{\text{JC}} = 2.0$	$2I_{\text{peak}}$	Historical JC-parity calibration	Legacy comparisons only
$s_{\text{JC}} = 1.0$	I_{peak}	Current validated Python/IPM conversion	Current map regeneration

Table 1: The two source-current conventions.

4 The two map conversions

Both paths use Eq. (2). They differ only in the extra source multiplier applied before the pump solve:

$$I_{\text{solve}} = s_{\text{JC}} I_{\text{peak}}, \quad s_{\text{JC}} = \text{pump-current-jc-scale}. \quad (7)$$

The factor-two path doubles the current amplitude. If interpreted as a matched-port power, that corresponds to a factor of four in power, or $10 \log_{10}(4) \simeq 6.02$ dB. It therefore moves the nonlinear operating point substantially and can make a gain ridge appear at an apparently unreachable external power.

The factor-one path does not add this historical JC multiplier. As of 18 July 2026 it is the validated conversion for regenerating the Python/IPM reference maps, and should be selected explicitly with

`--pump-current-jc-scale 1.0`

The command-line default remains 2.0 for backwards compatibility; relying on the default is consequently unsafe for new map regeneration.

5 Practical rule

Use the following checklist:

1. Convert dBm to I_{peak} using Eq. (2).
2. Do not apply an additional RMS/peak factor.
3. Use `-pump-current-jc-scale 1.0` for current Python/IPM maps.
4. Use 2.0 only when deliberately reproducing a historical JC-parity run.